No. Matrik

KEMENTERIAN PENDIDIKAN

BAHAGIAN MATRIKULASI
MATRICULATION DIVISION

PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI
MATRICULATION PROGRAMME EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

Kertas soalan ini mengandungi **11** halaman bercetak.
This question paper consists of **11** *printed pages.*

© Bahagian Matrikulasi

Answer **all** questions.
*Jawab **semua** soalan.*

1 (a) **FIGURE 1** shows the mass spectrum of zirconium, Zr which has 40 protons.

***RAJAH 1** menunjukkan spektrum jisim bagi zirkonium, Zr yang mempunyai 40 proton.*

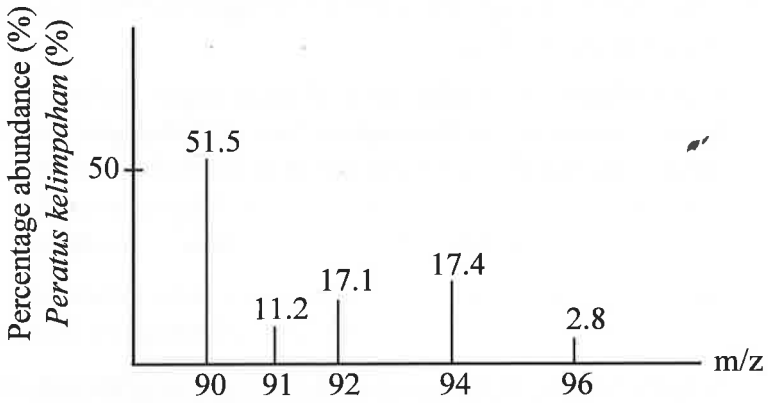


FIGURE 1
RAJAH 1

(i) Determine the relative atomic mass of zirconium.

Tentukan jisim atom relatif bagi zirkonium.

(ii) Write the isotopic notation when zirconium atom undergoes ionisation by losing four electrons.

Tuliskan notasi isotop apabila atom zirkonium mengalami pengionan dengan kehilangan empat elektron.

[4 *marks*]
[4 *markah*]

(b) An amount of 12.915 g of an organic compound containing carbon, hydrogen and oxygen was burned in excess oxygen. Analysis of the combustion revealed 18.942 g of carbon dioxide and 7.749 g of water were formed. Determine the empirical formula of the compound.

Sebanyak 12.915 g *sebatian organik mengandungi karbon, hidrogen dan oksigen telah dibakar dalam oksigen yang berlebihan. Analisis pembakaran menunjukkan pembentukan* 18.942 g *karbon dioksida dan* 7.749 g *air. Tentukan formula empirik sebatian tersebut.*

[4 *marks*]
[4 *markah*]

(c) A reagent bottle contains a concentrated sulphuric acid solution with 98% by mass and a density of 1.83 g mL^{-1}. Calculate the molarity of the acid.

Satu botol reagen mengandungi larutan asid sulfurik pekat dengan 98% *mengikut jisim dan ketumpatan* 1.83 g mL^{-1}. *Hitungkan kemolaran asid tersebut.*

[4 *marks*]
[4 *markah*]

(d) Potassium chlorate reacts with sulphur dioxide in an acidic medium according to the equation below. Balance the reaction equation by using ion-electron method.

Kalium klorat bertindak balas dengan sulfur dioksida dalam medium berasid mengikut persamaan di bawah. Imbangkan persamaan tindak balas tersebut dengan menggunakan kaedah ion-elektron.

$$ClO_3^- + SO_2 \rightarrow SO_4^{2-} + Cl^-$$

[3 *marks*]
[3 *markah*]

(e) Silicon nitride, Si_3N_4 is a very hard, high-temperature resistant ceramic used as component of turbine blades in jet engines. It is produced according to the equation below:

Silikon nitrida, Si_3N_4 *adalah seramik yang sangat keras, tahan suhu tinggi dan digunakan sebagai komponen bilah turbin dalam enjin jet. Ia dihasilkan mengikut persamaan di bawah*:

$$3Si(s) + 2N_2(g) \rightarrow Si_3N_4(s)$$

(i) Calculate the mass of Si_3N_4 produced when 20.00 g of Si and 15.00 g of N_2 are reacted.

Hitung jisim Si_3N_4 *yang terhasil apabila* 20.00 g Si *dan* 15.00 g N_2 *bertindak balas.*

(ii) Determine the percentage yield if 24.75 g of Si_3N_4 is obtained in this reaction.

Tentukan peratus hasil jika 24.75 g Si_3N_4 *diperolehi dalam tindak balas ini.*

[6 *marks*]
[6 *markah*]

2 (a) **FIGURE 2** shows a visible region of emission spectrum of hydrogen. The lines in this spectrum correspond to the electronic transitions.

RAJAH 2 *menunjukkan spektrum pemancaran hidrogen di kawasan nampak. Garis dalam spektrum tersebut adalah sepadan dengan peralihan elektron.*

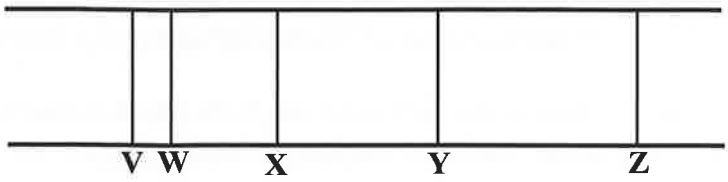


FIGURE 2
RAJAH 2

(i) Determine the line that corresponds to the shortest wavelength of radiation.

Tentukan garis yang sepadan dengan sinaran yang mempunyai panjang gelombang terpendek.

(ii) Determine the wavelength of the photon emitted (in nm) of line **X**.

Tentukan panjang gelombang pancaran foton (dalam nm*) bagi garis* **X**.

[5 *marks*]
[5 *markah*]

(b) The proton number of element **E** is 25.

Nombor proton bagi unsur **E** *adalah* 25.

(i) Write the electronic configuration of $\mathbf{E}^{2+}$ ion.

Tuliskan konfigurasi elektron bagi ion $\mathbf{E}^{2+}$.

(ii) Determine the number of unpaired electron(s) in **E**.

Tentukan bilangan elektron tak berpasangan dalam **E**.

(iii) Write a set of quantum numbers for the 9th electron in **E**.

Tuliskan satu set nombor kuantum bagi elektron ke-9 dalam **E**.

(iv) Draw the shape of the orbital occupied by the outermost electron(s) in **E**.

Lukiskan bentuk orbital yang diduduki oleh elektron di petala terluar bagi **E**.

[5 *marks*]
[5 *markah*]

3 (a) Boron trifluoride, BF_3 and iodine trifluoride, IF_3 are simple covalent molecules.

Boron triflourida, BF_3 *dan iodin trifluorida,* IF_3 *merupakan molekul kovalen ringkas.*

(i) Draw the Lewis structures for both molecules.

Lukiskan struktur Lewis bagi kedua-dua molekul.

(ii) Predict the molecular shapes for both molecules based on Valence Shell Electron Pair Repulsion theory.

Ramalkan bentuk molekul bagi kedua-dua molekul berdasarkan teori Penolakan Pasangan Elektron Petala Valens.

(iii) Differentiate the polarity of both molecules.

Bezakan kekutuban bagi kedua-dua molekul.

[8 *marks*]
[8 *markah*]

(b) Methanol, CH_3OH and pentane, $CH_3CH_2CH_2CH_2CH_3$ are colourless, volatile and flammable liquids.

Metanol, CH_3OH *dan pentana,* $CH_3CH_2CH_2CH_2CH_3$ *merupakan cecair tak berwarna, mudah meruap dan mudah terbakar.*

(i) Illustrate the hybridisation process of carbon and oxygen in CH_3OH.

Gambarkan proses penghibridan karbon dan oksigen dalam CH_3OH.

(ii) Draw and label the overlapping orbitals in CH_3OH.

Lukis dan labelkan pertindihan orbital dalam CH_3OH.

(iii) Compare the boiling points of CH_3OH and $CH_3CH_2CH_2CH_2CH_3$.

Bandingkan takat didih antara CH_3OH *dan* $CH_3CH_2CH_2CH_2CH_3$.

[9 *marks*]
[9 *markah*]

4 (a) Potassium chlorate, $KClO_3$ decomposes upon heating as below:

Kalium klorat, $KClO_3$ *terurai bila dipanaskan seperti di bawah*:

$$2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$$

When a sample of $KClO_3$ is heated, 240 mL of oxygen gas with pressure 755 torr is collected over water at 26°C. Calculate the mass of the decomposed $KClO_3$.
[Pressure of water vapor at 26°C is 25 torr]

Apabila satu sampel $KClO_3$ *dipanaskan*, 240 mL *gas oksigen bertekanan* 755 torr *terkumpul melalui sesaran air pada suhu* 26°C. *Hitung jisim* $KClO_3$ *yang terurai.*
[*Tekanan wap air pada* 26°C *ialah* 25 torr]

[5 *marks*]
[5 *markah*]

(b) Pentane, C_5H_{12} is a liquid whereas methane, CH_4 is a gas at room temperature. Explain the difference in physical state of both hydrocarbons based on intermolecular forces.

Pentana, C_5H_{12} *ialah cecair manakala metana*, CH_4 *ialah gas pada suhu bilik. Jelaskan perbezaan keadaan fizikal bagi kedua-dua hidrokarbon berdasarkan daya antara molekul.*

[4 *marks*]
[4 *markah*]

5 (a) The water-gas shift reaction is important in several chemical processes, such as the production of H_2 for fuel cells. The reaction can be written as follows:

Tindak balas anjakan air-gas adalah penting dalam beberapa proses kimia, seperti penghasilan H_2 *untuk sel bahan api. Tindak balas tersebut boleh ditulis seperti berikut*:

$$H_2(g) + CO_2(g) \rightleftharpoons H_2O(g) + CO(g) \qquad K_c = 0.106$$

A mixture of gases initially contains 0.025 M H_2 and 0.015 M CO_2 and is allowed to attain equilibrium at 700 K. Determine the equilibrium concentrations of H_2 and CO_2.

Satu campuran gas pada mulanya mengandungi 0.025 M H_2 *dan* 0.015 M CO_2, *dan dibiarkan mencapai keseimbangan pada suhu* 700 K. *Tentukan kepekatan* H_2 *dan* CO_2 *pada keseimbangan.*

[4 *marks*]
[4 *markah*]

(b) A mixture consisting of 0.42 M of N_2, 0.20 M of H_2 and 0.10 M of NH_3 is placed in a vessel at certain temperature and pressure, and allowed to react based on the following equation:

Satu campuran yang terdiri daripada 0.42 M N_2, 0.20 M H_2 *dan* 0.10 M NH_3 *diletakkan di dalam balang pada suhu dan tekanan tertentu, dan dibiarkan bertindak balas berdasarkan persamaan berikut*:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad K_c = 0.0584$$

(i) Determine the direction of the net reaction under these conditions.

Tentukan arah bagi tindak balas bersih pada keadaan ini.

(ii) Explain the effect on the equilibrium position when the volume of the vessel is decreased.

Jelaskan kesan terhadap kedudukan keseimbangan apabila isipadu bekas dikurangkan.

[5 *marks*]
[5 *markah*]

6 (a) Dimethylamine, $(CH_3)_2NH$ is a colourless liquid with an ammonia-like odour. The ionisation constant, K_b for $(CH_3)_2NH$ is 5.9×10^{-4}. Determine the pH of 1.5 M of $(CH_3)_2NH$ solution.

Dimetilamina, $(CH_3)_2NH$ *adalah cecair tak berwarna dengan bau seperti ammonia. Pemalar pengionan,* K_b *bagi* $(CH_3)_2NH$ *ialah* 5.9×10^{-4}. *Tentukan* pH *bagi larutan* $(CH_3)_2NH$ 1.5 M.

[5 *marks*]
[5 *markah*]

(b) A salt is produced when sodium hydroxide solution, NaOH reacts with hydrocyanic acid, HCN. Classify the salt formed according to its acid-base properties.

Suatu garam terhasil apabila larutan natrium hidroksida, NaOH *bertindak balas dengan asid hidrosianik,* HCN. *Kelaskan garam yang terbentuk mengikut sifat asid-bes.*

[3 *marks*]
[3 *markah*]

(c) The solubility product of magnesium phosphate, $Mg_3(PO_4)_2$ is 5.2×10^{-24}.

Hasil darab keterlarutan bagi magnesium fosfat, $Mg_3(PO_4)_2$ *ialah* 5.2×10^{-24}.

(i) Calculate the molar solubility of $Mg_3(PO_4)_2$.

Hitungkan keterlarutan molar bagi $Mg_3(PO_4)_2$.

(ii) Explain the effect on the molar solubility when magnesium chloride solution is added into the above salt solution.

Jelaskan kesan terhadap kelarutan molar apabila larutan magnesium klorida ditambah ke dalam larutan garam di atas.

[6 *marks*]
[6 *markah*]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT